Grand Rounds · October 2026

SIGN-OUT

Question 29

Anaphylaxis: PLC, IP3, Ca^{2+}

BLOCK	QUESTION	VIDEO
01	29	3 of 13

What you need to know before moving on.

Question 29

A 31-year-old man is brought to the emergency department 15 minutes after being stung by a bee while gardening. He has diffuse itching, hoarseness, and shortness of breath. He had a large local reaction to a bee sting 1 year ago.

Pulse is 128/min, respirations are 26/min, and blood pressure is 78/46 mm Hg. Examination shows generalized urticaria, swelling of the lips, and diffuse expiratory wheezes. Intramuscular epinephrine is administered.

Release of preformed histamine from this patient's mast cells was most directly triggered by which of the following intracellular events?

A	Adenylyl cyclase activation with increased cyclic AMP
B	Cyclooxygenase-mediated conversion of arachidonic acid
C	Phospholipase C activation with inositol trisphosphate–mediated calcium release
D	Janus kinase activation with STAT phosphorylation
E	Nitric oxide–induced increase in cyclic GMP

ANSWER AND VIDEO ON THE NEXT PAGE

QUESTION 29 · ANAPHYLAXIS: PLC, IP3, CA^{2+}

Answer C.

Phospholipase C activation with inositol trisphosphate–mediated calcium release

THE CLASS

25%

got it right (3 of 12)

MOST CHOSEN WRONG

B

4 of 12 · Cyclooxygenase-mediated conversion of arachidonic acid

Watch the explanation

Opens the video in your browser.

GRAND ROUNDS